

WEEK – 4

JAVA & MySQL DATABASE CONNECTIONS – SIMPLE – PREPARED – CALLABLE – THREE EXECUTION TYPES

Aim

To establish a connection between a Java application and a MySQL database using JDBC and perform database operations using Statement, PreparedStatement, and CallableStatement.

Software Requirements

- Ubuntu Linux
- Java JDK
- MySQL Server
- MySQL Connector/J (JDBC Driver)
- Terminal
- MySQL Client

Theory

Java Database Connectivity (JDBC) is an API that allows Java programs to communicate with relational databases. JDBC provides interfaces and classes to establish a database connection, execute SQL statements, retrieve results, and manage database resources.

The three major execution types used in JDBC are:

1. Statement

- Executes normal SQL queries.
- Used for static SQL.
- Suitable for simple SQL statements.

2. PreparedStatement

- Executes parameterized SQL queries.
- Prevents SQL Injection.
- Faster for repeated execution.

3. CallableStatement

- Executes stored procedures.
- Used when database logic is implemented inside MySQL

Database Creation

```
CREATE DATABASE college;
USE college;
CREATE TABLE student(
id INT PRIMARY KEY,
name VARCHAR(50)
);
INSERT INTO student VALUES
(101,'Ramesh'),
(102,'Haswika'),
(103,'Harshith');
```

```
mysql> CREATE DATABASE college;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> USE college;
Database changed
mysql>
mysql> CREATE TABLE student(
-> id INT PRIMARY KEY,
-> name VARCHAR(50)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> INSERT INTO student VALUES
-> (101,'Ramesh'),
-> (102,'Haswika'),
-> (103,'Harshith');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT * FROM student;^C
mysql> SELECT * FROM student;
+-----+-----+
| id  | name    |
+-----+-----+
| 101 | Ramesh  |
| 102 | Haswika |
| 103 | Harshith|
+-----+-----+
3 rows in set (0.00 sec)
```

**Download
JDBC
Driver**

Wget
<https://dev.mysql.com/get/Downloads/Connector-J/mysql-connector-j-9.4.0.tar.gz>

Extract
tar -xzf
mysql-
connector-j-

9.4.0.tar.gz

```
ttl@3@ITLAB3-1:~$ wget https://dev.mysql.com/get/Downloads/Connector-J/mysql-connector-j-9.4.0.tar.gz
--2026-07-14 15:06:29-- https://dev.mysql.com/get/Downloads/Connector-J/mysql-connector-j-9.4.0.tar.gz
Resolving dev.mysql.com (dev.mysql.com)... 23.9.116.144, 2600:1417:75:49d::2e31, 2600:1417:75:483::2e31
Connecting to dev.mysql.com (dev.mysql.com)|23.9.116.144|:443... connected.
HTTP request sent, awaiting response... 302 Moved Temporarily
Location: https://cdn.mysql.com/archives/mysql-connector-java-9.4/mysql-connector-j-9.4.0.tar.gz [following]
--2026-07-14 15:06:30-- https://cdn.mysql.com/archives/mysql-connector-java-9.4/mysql-connector-j-9.4.0.tar.gz
Resolving cdn.mysql.com (cdn.mysql.com)... 23.205.218.66, 2600:1417:75:489::1d68, 2600:1417:75:499::1d68
Connecting to cdn.mysql.com (cdn.mysql.com)|23.205.218.66|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4493403 (4.3M) [application/x-tar-gz]
Saving to: 'mysql-connector-j-9.4.0.tar.gz.1'

mysql-connector-j-9.4.0.tar.gz.1      100%[=====] 4.29M  --.-KB/s  in 0.1s

2026-07-14 15:06:30 (37.8 MB/s) - 'mysql-connector-j-9.4.0.tar.gz.1' saved [4493403/4493403]

ttl@3@ITLAB3-1:~$ tar -xzf mysql-connector-j-9.4.0.tar.gz
ttl@3@ITLAB3-1:~$ ls
24EU08143  Documents  folder_143  Music  mysql-connector-j-9.4.0.tar.gz  'New Folder 1'  Public  snap  Videos
Desktop    Downloads  folder_name mysql-connector-j-9.4.0  mysql-connector-j-9.4.0.tar.gz.1  Pictures  sample.cpp  Templates
```

Program – Statement

```
import java.sql.*;
public class JdbcDemo {
    public static void main(String args[]) {
        try {
            Connection con=DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/college",
                "root",
                "password");
            Statement stmt=con.createStatement();
            ResultSet rs=stmt.executeQuery("SELECT * FROM student");
            while(rs.next()){
                System.out.println(rs.getInt("id")+" "+rs.getString("name"));
            }
            con.close();
        }
        catch(Exception e){
            System.out.println(e);
        }
    }
}
```

Compilation

```
javac -cp ./~/mysql-connector-j-9.4.0/mysql-connector-j-9.4.0.jar JdbcDemo.java
```

Run

```
java -cp ./~/mysql-connector-j-9.4.0/mysql-connector-j-9.4.0.jar JdbcDemo
```

```
itlab3@ITLAB3-1:~$ nano JdbcDemo.java
itlab3@ITLAB3-1:~$ javac -cp ./mysql-connector-j-9.4.0/mysql-connector-j-9.4.0.jar JdbcDemo.java
itlab3@ITLAB3-1:~$ java -cp ./mysql-connector-j-9.4.0/mysql-connector-j-9.4.0.jar JdbcDemo

101 Ramesh
102 Haswika
103 Harshith
```

Output

Statement Output

101 Ramesh
102 Haswika
103 Harshith

Result

Thus, the Java application was successfully connected to the MySQL database using JDBC. SQL operations were performed successfully using Statement, PreparedStatement, and CallableStatement, and the expected results were obtained.